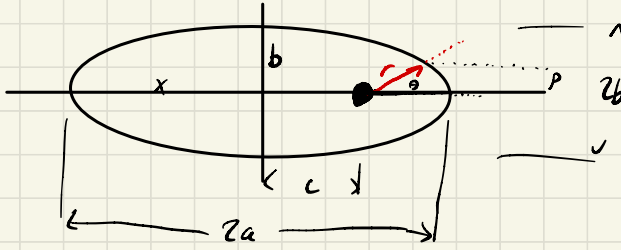


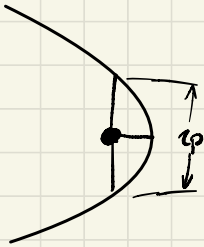
ellipse

$$0 < e < 1$$



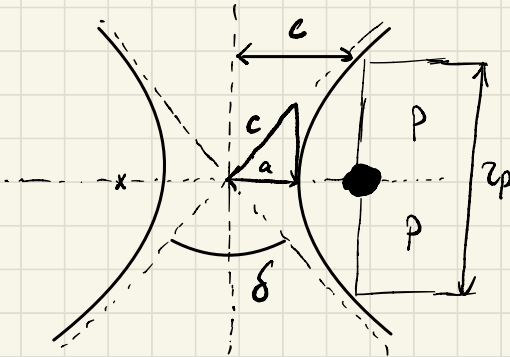
$$e = \frac{c}{a}$$

Parabola

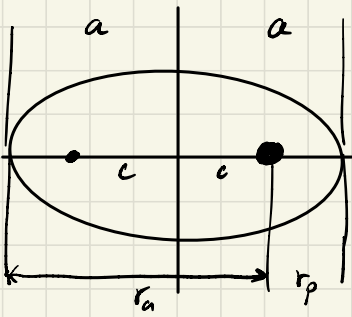


$$a = \infty$$

Hyperbola



Now let's look at the ellipse again



$$c + r_p = a$$

$$ea + r_p = a \rightarrow \begin{cases} r_p = a(1-e) \\ r_a = a(1+e) \end{cases}$$

Also...

$$r_a + r_p = 2a \quad r_a - r_p = 2c \quad e = \frac{c}{a} = \frac{2c}{2a} = \frac{r_a - r_p}{r_a + r_p} = e$$

and...

$$r = \frac{p}{1 + e \cos \theta}$$

periapsis $\theta = 0^\circ$
apoapsis $\theta = 180^\circ$

$$r_p = \frac{p}{1+e}$$

$$r_a = \frac{p}{1-e}$$

Finally...

$$r_p = \frac{p}{1+e} = a(1-e) \rightarrow p = a(1-e)(1+e)$$

$$p = a(1-e^2)$$

We can also relate specific energy to the semi-major axis:

At periapsis...

$$\begin{aligned} \epsilon &= \frac{v_p^2}{2} - \frac{\mu}{r_p} = \frac{v_p^2 r_p^2}{2r_p^2} - \frac{\mu r_p}{r_p^2} \\ &= \frac{v_p^2 r_p^2 - \mu r_p}{2r_p^2} = \frac{h^2 - \mu r_p}{2r_p^2} \end{aligned}$$

$h = r_p v_p \cos \delta$

$$p = \frac{h^2}{\mu} \rightarrow h^2 = p\mu$$

$$\epsilon = \frac{\mu p - 2\mu r_p}{2r_p^2} = \mu \left(\frac{p - 2r_p}{2r_p^2} \right) \quad p = r_p + er_p$$

$$= \mu \left(\frac{r_p + er_p - 2r_p}{2r_p^2} \right) = \mu \left(\frac{r_p(e-1)}{2r_p^2} \right)$$

$$\epsilon = \mu \left(\frac{e-1}{2r_p} \right) = -\mu \left(\frac{1-e}{2r_p} \right)$$

recall that $r_p = a(1-e) \rightarrow$ substitute

$$\epsilon = -\mu \left(\frac{1-e}{2a(1-e)} \right) = -\frac{\mu}{2a} \quad \text{Thus, } \boxed{\epsilon = -\frac{\mu}{2a}}$$

lets take a look at the specific angular momentum.

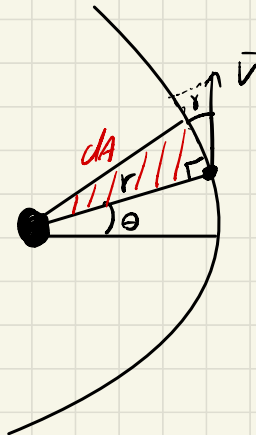
$$h = r v \cos(\gamma)$$

$$\text{Then, } h = r r \dot{\theta} = r^2 \frac{d\theta}{dt}$$

$$h = 2 \frac{dA}{dt}$$

$$dt = \frac{2}{h} dA$$

Kepler's Second Law



$$v \cos \gamma = r \dot{\theta}$$

$$dA = \frac{1}{2} r r d\theta$$

$$2dA = r^2 d\theta$$

integrate...

$$\int dt = \int \frac{2}{h} dA$$

$$P = \frac{2}{h} A$$

Period

Area of an ellipse: $A = \pi ab$

$$b = \sqrt{a^2 - c^2}$$

$$c = ae$$

$$b = \sqrt{a^2 - a^2 e^2}$$

$$b = \sqrt{a^2 (1 - e^2)}$$

$$b = \sqrt{ap}$$

$$\text{Thus... } P = \frac{2\pi a \sqrt{ap}}{h}$$

$$P = \frac{2\pi a^{3/2} \sqrt{p}}{h}$$

$$h = \sqrt{\mu p}$$

$$\frac{\sqrt{p}}{h} = \frac{1}{\sqrt{\mu}}$$

$$P = 2\pi \sqrt{\frac{a^3}{\mu}}$$

Period of orbit - Kepler's Third Law

Note: Proof could show up on a test!

Velocity for a circular orbit

$$e=0 \quad r=p=a$$

$$E = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} \rightarrow \frac{v^2}{2} - \frac{\mu}{a} = -\frac{\mu}{2a} \rightarrow \frac{v^2}{2} = \frac{\mu}{2a} - \frac{\mu}{2a}$$

$$\text{Thus... } v_c = \sqrt{\frac{\mu}{a}} = \sqrt{\frac{\mu}{r}}$$

Example: The ISS is in a $\sim 410 \text{ km}$ circular orbit about earth.
What is the orbital speed, specific energy, period, and spec. ang. mom.

A circular orbit $e=0$ $r_c = a = p = \text{const.}$

$$r_c = \underbrace{6378 \text{ km}}_{r_{\text{earth}}} + 410 \text{ km} = 6788 \text{ km} \quad \mu = 3.986 \times 10^5 \frac{\text{km}^3}{\text{s}^2}$$

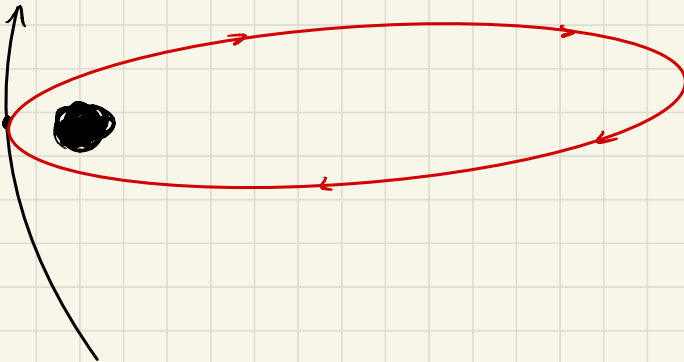
Speed: $v_c = \sqrt{\frac{\mu}{r}} = 7.663 \frac{\text{km}}{\text{s}}$

Specific energy: $\xi = -\frac{\mu}{2r_c} = -29.36 \frac{\text{km}^2}{\text{s}^2}$

Period: $P = 2\pi \sqrt{\frac{a^3}{\mu}} = 5,566 \text{ sec}$ or 92.76 min

Specific ang. mom. $h = r v \cos \delta = r_c v_c = 5.2 \times 10^4 \frac{\text{km}^2}{\text{s}}$

Example on march 10th, 2006 the Mars Reconnaissance Orbiter (MRO) performed a propulsive maneuver to insert into a highly elliptical orbit around Mars. This was the beginning of a period of aerobraking. This initial orbit had an apoapsis of $45,000 \text{ km}$ and periapsis at 450 km .
What is the orbit semi-major axis, ecc., period, and speed...



Compute radii: $r_a = 3,389 \text{ km} + 45,000 = 48,389 \text{ km}$
 $r_p = 3,389 \text{ km} + 450 = 3,819 \text{ km}$

Semi-major: $a = \frac{r_a + r_p}{2} = 26,104 \text{ km}$

eccentricity: $e = \frac{r_a - r_p}{r_a + r_p} = 0.85$

Period: $P = 2\pi \sqrt{\frac{a^3}{\mu}} = 1.277 \times 10^5 \text{ sec} = 35.48 \text{ hr}$

Speed at periapsis: Method #1: $E = -\frac{\mu}{2a} = \frac{v_p^2}{2} - \frac{\mu}{r_p} \rightarrow v_p = 4.57 \frac{\text{km}}{\text{s}}$

Method #2: $P = r_p(1+e)$
 $h = \sqrt{\mu r_p} = v_p r_p$ $v_p = \sqrt{\frac{\mu(1+e)}{r_p}} = 4.57 \frac{\text{km}}{\text{s}}$

We have just considered examples of circular and elliptical orbits.

- If we imagine an elliptical orbit which apsis is stretched further until reaching infinity

$$E = -\frac{\mu}{2a} = -\frac{\mu}{2\infty} = 0$$

more over...

$$E = \frac{v^2}{2} - \frac{\mu}{r} = 0$$

as $r \rightarrow \infty \dots \mathcal{E} = \frac{v^2}{2} \rightarrow 0$ thus, $v \rightarrow \infty$

$v = 0$ @ r_∞ for a parabolic orbit

$$\frac{v^2}{2} - \frac{\mu}{r} = 0 \rightarrow \frac{v^2}{2} = \frac{\mu}{r}$$

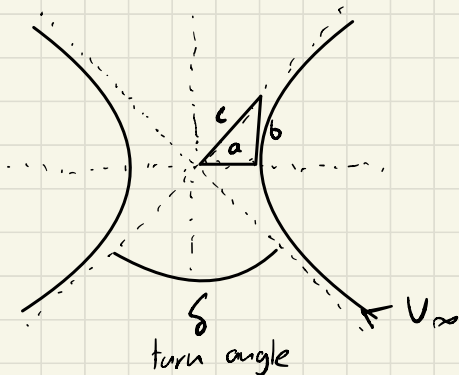
$$v_{esc} = \sqrt{\frac{2\mu}{r}} \rightarrow \text{escape velocity}$$

If $v < v_{esc} \rightarrow \mathcal{E} < 0 \rightarrow$ closed orbit

$v > v_{esc} \rightarrow \mathcal{E} > 0 \rightarrow$ open orbit

*Note... we rarely encounter purely parabolic orbits

If we reach r_∞ with some velocity, then we must be in a hyperbolic orbit...



$$\sin\left(\frac{\delta}{2}\right) = \frac{a}{c}$$

$$e = \sqrt{1 + \frac{b^2}{a^2}}$$

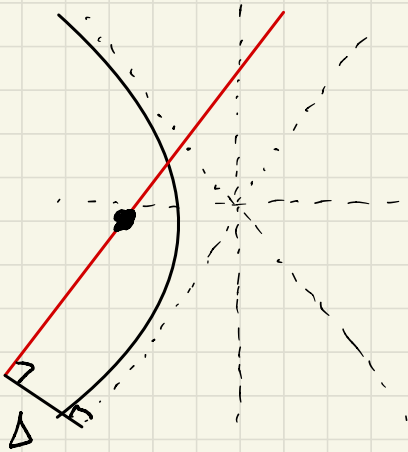
$$a^2 + c^2 = a^2 + b^2 = c^2$$

Thus, $c = ae$

$$\text{and } \sin\left(\frac{\xi}{2}\right) = \frac{a}{ae} = \frac{1}{e}$$

Consider the speed far away from the central body

$$\xi = \frac{V^2}{2} - \frac{\mu}{r} = \frac{V_\infty^2}{2} - \frac{\mu}{\infty} = \frac{V_\infty^2}{2} \rightarrow \xi = \frac{V_\infty^2}{2} > 0$$



Δ = miss distance

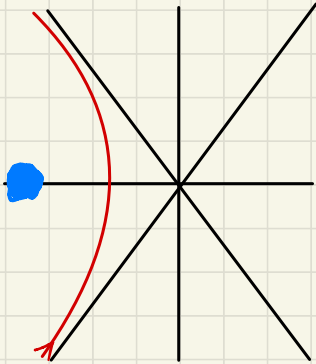
$$\bar{h} = \bar{r} \times \bar{v}$$

$$= V_\infty \Delta \hat{k}$$

$$h = V_\infty \Delta$$

Example: Consider a spacecraft performing a flyby of Venus, on approach it has a hyperbolic excess speed of $V_{\infty} = 3.5 \text{ km/s}$ and a miss distance of $b = 30,000 \text{ km}$.

Find specific energy, specific angular momentum, periapsis radius, and turn angle.



$$\mathcal{E} = \frac{V_{\infty}^2}{2} = 6.125 \frac{\text{km}^2}{\text{s}^2}$$

$$h = V_{\infty} b = 105,000 \frac{\text{km}^2}{\text{s}}$$

$$\mu = 3.257 \times 10^5 \frac{\text{km}^3}{\text{s}^2}$$

$$\mathcal{E} = -\frac{\mu}{2a} \rightarrow a = -\frac{\mu}{2\mathcal{E}} = -2.66 \times 10^4 \text{ km}$$

$$p = \frac{h^2}{\mu} = a(1 - e^2) = 3.385 \times 10^4 \text{ km}$$

$$\downarrow$$

$$e = \sqrt{1 - \frac{p}{a}} \rightarrow e = 1.508$$

$$r_p = a(1 - e) = 13,500 \text{ km}$$

$$\sin\left(\frac{\delta}{2}\right) = \frac{1}{e} \rightarrow \delta = 2 \sin^{-1}\left(\frac{1}{e}\right) = 83 \text{ deg}$$

Orbital elements...

Two body orbits follow the path of a conic section.
This is a planar trajectory through space

We can define the shape and orientation of a two-body conic using the orbital elements

6 classical orbital elements that describe the shape and orientation of an orbit relative to a convenient reference frame

- Let $\{\hat{i}, \hat{j}, \hat{k}\}$ be the basis for this inertial frame.

a : semi-major axis (size of orbit)

e : eccentricity (shape)

i : inclination - angle between $\hat{k}$ and the orbit's $\hat{n}$.

Ω : right ascension of the ascending Node (RAAN) - angle $\hat{i}$ - $\hat{j}$ plane between $\hat{i}$ and where the orbit passes from negative to positive z -values.

ω : argument of pericapsis - angle in the plane of the orbit between the ascending node and the pericapsis (Measured in the direction of orbit)

T : time of pericapsis passage (Also called true anomaly at epoch)

Reference Frames

The most important inertial frame for astrodynamics is:

ICRF (Inertial Celestial Reference Frame)

The international celestial reference system is ICRS

* we define ICRF from ICRS

The origin is at the Solar system's barycenter (SSB)
and its axes are fixed relative to distant objects (quasars)

↳ reference sources are the objects in space

ICRF (Realization) of ICRS (specification)

The alignment of ICRS axes was chosen to be such that:

"The principle plane should be as near as possible to the mean equator at J200.0 and the origin in this principle plane be as near to dynamical equinox at J200"

↓

z-axis (defines the principle plane): perpendicular to Earth's equatorial plane

x-axis: ascending node of the ecliptic plane at equatorial plane

y-axis: complete R.H.S

While the axis are fixed to distant sources, the initial alignment was selected based on the following dynamical frame at J2000 (No longer perfectly aligned)

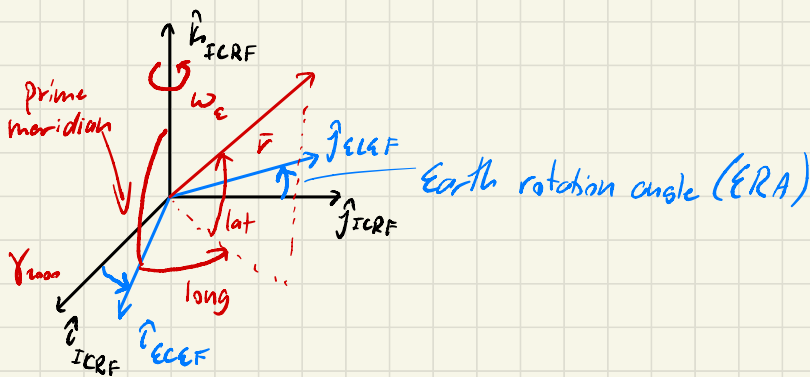
Important: Since both planes (equatorial and elliptical) are moving, you must specify an epoch at which the frame was generated

ICRF specifies the "usual" direction of the coordinate axes for inertial frames used in spacecraft and astronomy

we say things like Earth-Centered Inertial (ECI) ... etc

↳ We usually mean that the ICRF axes are moved of the specified central body.

The other common frame is Earth-Centered-Earth-Fixed (ECEF), with origin at center of earth and coordinate axis rigidly fixed.



$$\text{ERA} = \text{GMST} + \epsilon_{\text{prec}} \quad \leftarrow \text{precession of equinoxes}$$



Greenwich mean Sidereal time (Angle) (15 degrees per hour)

Solar and Sidereal time

Solar time: based on the time for the sun to return to a reference meridian

↓ in mean solar time: 1 day = 24 hrs = 86,400s

length of our SI second is 9,192,631,770 periods of radiation of Cesium-133.



Selected for 86,400 sec day in the year 1900.

Sidereal time:

based on time to return equinox to reference meridian

Atomic time and civil time

Int. Atomic time (TAI): Continuous time scale based on the SI second

Civil time (UTC): Adjustment of TAI by an integer number of leap seconds to keep UTC within 0.9 sec of UT1 (based on ERA)

Canonical units

Canonical units normalize "regular units" (eg. km, $\frac{1}{s}$) to quantities that are of a similar magnitude and sometimes lead to better numerical performance when working in limited precision.

Procedure:

1. Pick a reference distance: distance unit (DU)

a. earth orbit $1\text{DU} = R_e = 6378\text{ km}$

b. Sun orbit $1\text{DU} = 1\text{AU} = 1.496 \times 10^8\text{ km}$

2. Define $\mu = \frac{\text{DU}^3}{\text{TU}^2} \leftarrow$ This sets TU

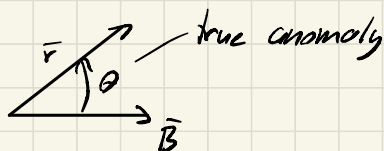
3. Compute TU $\rightarrow 1\text{TU} = \sqrt{\frac{1\text{DU}^3}{\mu}}$ eg. earth: $\text{TU} = 807\text{ sec}$

The perifocal frame

The perifocal frame is one of the most common frames we will use this semester.

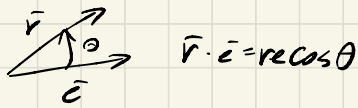
↳ origin is at the focus (location of central body), with x-axis pointing towards the periapsis.

Recall from before:



$$\begin{aligned}\vec{r} \cdot \vec{B} &= rB \cos \theta \\ \vec{B} &= \vec{r} \times \hat{h} = \mu \frac{\vec{r}}{r}\end{aligned}$$

then we said $\mathbf{e} = \frac{\mathbf{B}}{h} \rightarrow \bar{\mathbf{e}} = \frac{1}{h} \bar{\mathbf{B}} = \frac{1}{h} \left(\bar{\mathbf{r}} \times \bar{\mathbf{h}} - \frac{\bar{\mathbf{r}}}{r} \right)$



$$\bar{\mathbf{r}} \cdot \bar{\mathbf{e}} = r \cos \theta$$

and we observed that periaapsis occurs at $\theta = 0^\circ$

★ IMPORTANT: STUDY THIS

therefore, $\bar{\mathbf{e}}$ points from the focus towards periaapsis and is the x-axis.

Define $\hat{\mathbf{p}}$ as the unit vector in the perifocal frame in the x-direction

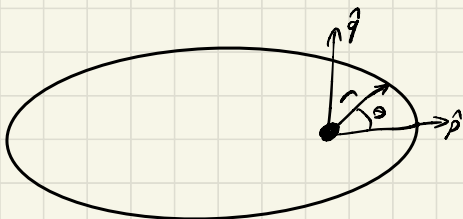
$$\hat{\mathbf{p}} = \frac{\bar{\mathbf{e}}}{\|\bar{\mathbf{e}}\|} = \frac{\bar{\mathbf{e}}}{e}$$

Define $\hat{\mathbf{w}}$ as normal to the orbit plane, positive in the direction of the specific angular momentum.

$$\hat{\mathbf{w}} = \frac{\bar{\mathbf{h}}}{\|\bar{\mathbf{h}}\|} = \frac{\bar{\mathbf{h}}}{h}$$

let $\hat{\mathbf{q}}$ complete the right hand system (RHS)

$$\hat{\mathbf{q}} = \hat{\mathbf{w}} \times \hat{\mathbf{p}}$$



$$\bar{\mathbf{r}} = x\hat{\mathbf{p}} + y\hat{\mathbf{q}}$$

$$x = r \cos \theta, y = r \sin \theta, z = 0$$

$$R_I^P = [\hat{p}, \hat{q}, \hat{\omega}] \rightarrow \text{from } P \text{ to } I$$

$$R_P^I = \begin{bmatrix} \hat{p}^T \\ \hat{q}^T \\ \hat{\omega}^T \end{bmatrix} \rightarrow \text{from } I \text{ to } P$$